

Sonifying the Heart: Inner Emotion Detection from ECG through Music

Music Workshop Sprint 3

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1 The Project

In Sprint 1 we proposed turning ECG recordings into music, so that a healthy and an irregular heart rhythm sound different. In Sprint 2 we planned the pipeline.

In Sprint 3 we built it, and added a fourth stage. That stage reads the emotion of the generated music, as a window into the person’s inner state:

ECG data → Heartbeats → Music → Emotion

The code is a Python package, `ecgmusic`, with a step-by-step notebook, `main.ipynb`. One command turns a PhysioNet record, or any ECG file, into MIDI, audio, a score plot and an emotion estimate.

2 What We Built

- **ECG data.**
 - The main source is the MIT-BIH Atrial Fibrillation Database (`afdb`). Most of its recordings contain both normal rhythm and AFib, so the same person is heard in both states.
 - `mitdb` adds records 201 and 202, which come from one patient.
 - Records are streamed from PhysioNet. Any WFDB or CSV file also works.
- **Heartbeats.**
 - R-peaks come from the annotations, or from the XQRS detector for other files. From them we compute the R–R intervals and RMSSD.
 - Each recording gives a 30 s normal clip (the steadiest window) and a 30 s AFib clip (the best-detected window). We choose windows this way because `afdb`’s automatic beat marks jitter in places.
- **Music.**
 - *Notes*: one per heartbeat. Each starts at the R-peak, lasts 85% of the R–R interval, and is as loud as the ECG amplitude at that beat.
 - *Pitch*: follows the deviation d_i of the interval from the mean of the last five intervals. It is C4 when the beat is on time, higher for a faster beat and lower for a slower one.
 - *Scale*: C-major pentatonic while $|d_i| < 12\%$, chromatic beyond that.
 - *Harmony*: strings play C–Am–F–G. Each chord becomes diminished ($\geq 12\%$) or a cluster ($\geq 25\%$) as the beats get irregular.
 - *Pulse*: a soft “lub-dub” drum beat follows every heartbeat.
- **Emotion.**

- Our model places the melody on Russell’s valence–arousal plane, using three cues: rhythmic irregularity, pitch spread and consonance.
- The emotion picks the melody instrument: nylon guitar for calm, distortion guitar for tense, marimba for happy and cello for sad.
- Two pretrained models, Essentia and music2emo, give independent opinions.

3 Results

We made 14 pieces: a normal and an AFib clip from each of 7 recordings of 6 people (Table 1).¹

Table 1: Every clip. Chords: consonant / tense / chaotic. Emotion from our model.

Recording	Rhythm	RMSSD (ms)	Off-scale notes	Chords	Emotion (valence, arousal)
afdb 04015	normal	7.6	0%	10 / 0 / 0	calm (+1.00, −0.97)
	AFib	126.1	57%	5 / 9 / 3	tense (−0.67, +0.76)
afdb 04043	normal	8.5	0%	14 / 0 / 0	calm (+1.00, −0.96)
	AFib	134.9	52%	6 / 10 / 0	tense (−0.51, +0.45)
afdb 04126	normal	18.6	0%	10 / 0 / 0	calm (+0.93, −0.79)
	AFib	141.9	66%	3 / 12 / 5	tense (−0.89, +0.85)
afdb 06995	normal	17.8	0%	10 / 0 / 0	calm (+0.93, −0.81)
	AFib	122.8	51%	6 / 7 / 1	tense (−0.45, +0.31)
afdb 08455	normal	8.3	0%	10 / 0 / 0	calm (+1.00, −0.96)
	AFib	99.9	31%	11 / 4 / 0	happy, borderline (+0.07, +0.02)
mitdb 201	normal	64.1	0%	6 / 0 / 0	calm (+0.89, −0.69)
	AFib	194.3	52%	3 / 9 / 1	tense (−0.58, +0.76)
mitdb 202	normal	37.4	0%	7 / 0 / 0	calm (+0.89, −0.65)
	AFib	146.9	33%	9 / 5 / 2	tense (−0.07, +0.44)

- **Heartbeats.** In every recording, the AFib clip is 3–17 times as variable as the normal clip.
- **Music.**
 - Normal pieces stay on the scale, and all 67 of their chords are consonant.
 - AFib pieces leave the scale for 31–66% of their notes, and 68 of their 111 chords turn tense or chaotic.
- **Emotion.** Our model reads all 7 normal pieces as calm, and 6 of the 7 AFib pieces as tense. The seventh, 08455, is a mild AFib stretch and lands at the centre of the plane.
- **Pretrained models.** Both hear a difference in every recording, but describe it differently:
 - Essentia hears the AFib pieces as more energetic and more pleasant (7 of 7).
 - music2emo hears them as less pleasant and less energetic (6 of 7).

Tempo probably explains part of this. Every AFib clip is also the faster heart (101–160 against 50–108 bpm), and our model ignores tempo on purpose.

4 Changes from the Sprint 2 Plan

- **Data source.** afdb is the main source, instead of mitdb with nsrdb as the reference. This lets us hear the same person in both rhythms.
- **Clip selection.** Choosing the steadiest window replaced the 0.3–2.0s plausibility filter. It handles the jittery beat marks we found.
- **Scale and loudness.**

¹These numbers come from our first implementation. The `ecgmusic` package uses the same rules and parameters, and will regenerate them.

- The scale is C-major pentatonic, which has the same notes as A minor pentatonic, to fit the C–Am–F–G harmony.
- The ECG amplitude sets the loudness. $|d_i|$ drives the harmony instead.
- **Added:** the three-part arrangement, the emotion stage, and support for any ECG file.
- **Not done yet:** SDNN, pNN50, Poincaré plots and rate-matched clips.

5 Next Steps

- Regenerate the results with the `ecgmusic` package.
- Test whether the emotion matches what people actually feel. We plan to use ECG datasets with self-reported emotion (WESAD, DREAMER, AMIGOS) and a listening study.
- Adapt the mapping for healthy hearts. There, calm breathing *raises* beat-to-beat variability, which our current mapping would read as tension.

6 Contributions

Chervith Reddy	2024101076	Stages 1–2: ECG data and heartbeats
Manidhar Sukasi	2023101067	Stage 3: music, and integration (package, notebook)
Sathwik Reddy	2024121002	Stage 4: emotion and evaluation